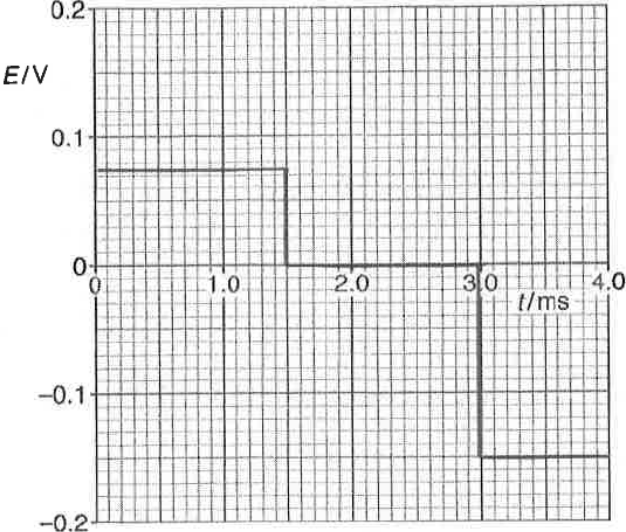


	The resistance wire should be clearly distinguishable from the connecting wires.
6(a)(i)	$B = \mu_0 n I$ $= \left(4\pi \times 10^{-7}\right) \left(\frac{900}{0.46}\right) (2.4)$ $= 0.00590$ $= 5.90 \times 10^{-3} \text{ T}$ Note: n is for number of turns per unit length of the solenoid.
6(a)(ii)	The flux at the centre of the solenoid are parallel and uniformly spaced out and are contributed by both halves of the solenoid. The flux at the end of solenoid spread out non-uniformly as they enter into free air and are only contributed by one half of the solenoid. So the magnitude of the flux density at the end will be <u>half</u> of that at the centre of the solenoid.
6(b)(i)	Faraday's law states that the magnitude of the induced e.m.f. in a <u>coil</u> is directly proportional to the <u>rate of change of magnetic flux linkage through that coil</u> .

No.	Solution
6(b)(ii)	$\Phi = NBA$ $E = \left -\frac{d\Phi}{dt} \right = \frac{\Delta\Phi}{\Delta t} = \frac{(95)(0.00590)(\pi \times 0.008^2)}{1.5 \times 10^{-3}}$ $= 0.0751 \text{ V}$
6(b)(iii)	 From $t = 3.0 \text{ ms}$ to 4.0 ms, $\Phi = NBA$ $= N(\mu_0 n I) A$ $E = \frac{\Delta\Phi}{\Delta t}$ $= \frac{(95)(4\pi \times 10^{-7})\left(\frac{900}{0.46}\right)(3.2)(\pi \times 0.008^2)}{1.0 \times 10^{-3}}$ $= 0.150 \text{ V}$
7(a)	The root mean square value of an alternating current is the equivalent value of a